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DETAIL SPECIFICATION

SPECTROMETRIC OIL STANDARDS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for blended spectrometric oil standards for use in calibrating or verifying the calibration of rotating disc, atomic emission spectrometers (RDE-AES) used in spectrometric analysis of metallic elements found in oils and other fluids.

1.2 Classification. The blended spectrometric oil standards are of the types shown in Table I.

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, B120-3, Highway 547, Lakehurst, NJ 08733-5100 or emailed to NAVAIR-Standardization@us.navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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Table I. Spectrometric oil standard types and uses.

Type	Nominal Concentrations Available (ppm)	Description	Use (Spectrometer Type (as described in 6.1)
D3	100	Spectrometric oil standard with 3 elements (Boron, Molybdenum, Zinc)	Daily Check (15, 30) Complete Standardization (15, 30) Integrated Standardization (30) Calibration (30)
D12	5, 10, 30, 50, 100, 300	Spectrometric oil standards with 12 elements (Aluminum, Chromium, Copper, Iron, Lead, Magnesium, Nickel, Silicon, Silver, Sodium, Tin, Titanium)	Daily Check (15, 30) Complete Standardization (15, 30) Integrated Standardization (30) Calibration (30)
D15X	5, 10, 30, 50, 100	Spectrometric oil standard with 15 elements (Arsenic, Barium, Bismuth, Cadmium, Calcium, Cerium, Cobalt, Indium, Lithium, Manganese, Phosphorus, Potassium, Vanadium, Tungsten, Zirconium)	Daily Check (30) Integrated Standardization (30) Calibration (30)
D19-0	0	Spectrometric oil standard with less than 1 ppm of 19 elements (Aluminum, Barium, Boron, Cadmium, Chromium, Copper, Iron, Lead, Magnesium, Manganese, Molybdenum, Nickel, Silicon, Silver, Sodium, Tin, Titanium, Vanadium, Zinc)	Daily Check (15, 30) Complete Standardization (15, 30) Calibration (15)
D19	5, 10, 30, 50, 80, 100, 120, 300, 500, 700, 900	Spectrometric oil standards with 19 elements (Aluminum, Barium, Boron, Cadmium, Chromium, Copper, Iron, Lead, Magnesium, Manganese, Molybdenum, Nickel, Silicon, Silver, Sodium, Tin, Titanium, Vanadium, Zinc)	Calibration (15, 30)
D30-0	0	Spectrometric oil standard with less than 1 ppm of 30 elements (Aluminum, Arsenic, Barium, Bismuth, Boron, Cadmium, Calcium, Cerium, Chromium, Cobalt, Copper, Indium, Iron, Lead, Lithium, Magnesium, Manganese, Molybdenum, Nickel, Phosphorus, Potassium, Silicon, Silver, Sodium, Tin, Titanium, Tungsten, Vanadium, Zinc, Zirconium)	Integrated Standardization (30) Calibration (30)
C24-100	100	Spectrometric oil standard with 24 elements (Aluminum, Antimony, Barium, Boron, Cadmium, Calcium, Chromium, Copper, Iron, Lead, Lithium, Magnesium, Manganese, Molybdenum, Nickel, Phosphorus, Potassium, Silicon, Silver, Sodium, Tin, Titanium, Vanadium, Zinc)	Integrated Standardization (30)

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2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-791	-	Lubricant, Liquid Fuel and Related Products, Methods of Testing.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

ASTM INTERNATIONAL

ASTM D92	-	Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester
ASTM D97	-	Standard Test Method for Pour Point of Petroleum Products
ASTM D445	-	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D2270	-	Standard Practice for Calculating Viscosity Index from Kinematic Viscosity at 40 ºC and 100 ºC
ASTM D4052	-	Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter
ASTM D4057	-	Standard Practice for Manual Sampling of Petroleum and Petroleum Products
ASTM D4177	-	Standard Practice for Automatic Sampling of Petroleum and Petroleum Products
ASTM D5185	-	Standard Test Method for Multielement Determination of Used and Unused Lubricating Oils and Base Oils by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES)
ASTM D6595	-	Standard Test Method for Determination of Wear Metals and Contaminants in Used Lubricating Oils or Used Hydraulic Fluids by Rotating Disc Electrode Atomic Emission Spectrometry

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ASTM D7042 - Standard Test Method for Dynamic Viscosity and Density of Liquids by Stabinger Viscometer (and the Calculation of Kinematic Viscosity)

(Copies of these documents are available online at https://www.astm.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org.)

SAE INTERNATIONAL

SAE J1899 - Lubricating Oil, Aircraft Piston Engine (Ashless Dispersant)
SAE J1966 - Lubricating Oils, Aircraft Piston Engine (Non-Dispersant)

(Copies of these documents are available from https://www.sae.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The spectrometric oil standards furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Material. The standards shall be prepared from metallo-organic concentrates and base oil.

3.2.1 Metallo-organic concentrates. The metallo-organic concentrates shall consist of metallo-organic or organic oil-miscible liquid compounds or solutions and a fully compatible preservative, stabilizing additive. The compounds shall be fully soluble/miscible in the base oil.

3.2.2 Base oil. The standards shall be prepared using the specified base oils outlined in 3.2.2.1 and 3.2.2.2.

3.2.2.1 D-Series base oil. The base oil for blending the standards shall be an SAE 50 single grade mineral lubricating oil produced from refined petroleum products that meets the requirements of table II. The base oil shall be tested to verify that no trace metals exceed 1 part per million (ppm), milligrams per kilogram of the elements outlined in 4.4.

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3.2.2.1.1 SAE J1966 and SAE J1899 engine oils. In the past, SAE J1966 non-additive mineral aircraft engine lubricating oil, or SAE J1899 ashless dispersant aircraft engine lubricating oil, have been used to meet the D-Series base oil requirement. These oils may contain pour point depressants up to a maximum amount of 1.0 percent by weight as well as an antioxidant in an amount not to exceed 0.5 percent by weight. Silicone antifoam additives shall not be used.

Table II. Chemical/physical property requirements and test methods for D-Series base oil.

Chemical / Physical Property	Requirement	Test Method
Density at 60 °F	1/	ASTM D4052
Viscosity (mm^2 /s at 100 °C)	16.3 min., 21.9 max.	ASTM D445 ASTM D7042
Viscosity (mm^2 /s at 40 °C)	1/	ASTM D445 ASTM D7042
Viscosity Index, Min	85	ASTM D2270
Trace Sediment	0.005 ml/100 mL Oil max.	FED-STD-791, Method 3044.7
Pour point	-12 °C max.	ASTM D97
Flash point, minimum (COC)	243 °C	ASTM D92
Metal Elements (RDE-AES)	Meets accuracy and repeatability as outlined in paragraph 4.4 for the elements in D-Series Base Oil / D19-0 and D-Series Base Oil / D30-0 respectively	ASTM D6595
Metal Elements (ICP-AES)	Meets accuracy and repeatability as outlined in paragraph 4.4 for the elements in D-Series Base Oil / D19-0 and D-Series Base Oil / D30-0 respectively	ASTM D5185

1/ Values are required and used as comparative reference for testing.

3.2.2.2 C-Series base oil. The base oil for blending the C-Series standards shall be a 75cSt white mineral lubricating oil produced from refined petroleum products that meets the requirements of table III. The base oil shall be tested to verify that no trace metals exceed 1 part per million (ppm), milligrams per kilogram of the elements outlined in 4.4.

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Table III. Chemical/physical property requirements and test methods for C-Series base oil.

Chemical / Physical Property	Requirement	Test Method
Density at 60 °F	1/	ASTM D4052
Viscosity (mm^2 /s at 100 °C)	10.5 max.	ASTM D445 ASTM D7042
Viscosity (mm^2 /s at 40 °C)	65.0 min., 80.0 max.	ASTM D445 ASTM D7042
Viscosity Index, Min	90	ASTM D2270
Trace Sediment	0.005 ml/100 mL Oil max.	FED-STD-791, Method 3044.7
Pour point	-15 °C max.	ASTM D97
Flash point, minimum (COC)	215 °C	ASTM D92
Metal Elements (RDE-AES)	Meets accuracy and repeatability as outlined in paragraph 4.4 for the elements in C-Series Base Oil	ASTM D6595
Metal Elements (ICP-AES)	Meets accuracy and repeatability as outlined in paragraph 4.4 for the elements in C-Series Base Oil	ASTM D5185

1/ Values are required and used as comparative reference for testing.

3.3 Spectrometric oil standards. The spectrometric oil standards shall be prepared using materials specified in 3.2 to the concentration specified in the contract (see 6.2). Nominal concentrations are as shown in table I.

3.3.1 Type D3-XXX. Type D3-XXX spectrometric oil standard shall meet the Metal Elements (RDE-AES and ICP-AES) requirements specified in table II and contain the elements at the concentration as listed in table IV.

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Table IV. List of elements and formulations by type.

Element (chemical symbol)	Spectrometric Oil Standards Types							
	D3-XXX	D12-XXX	D15X-XXX	D-Series Base Oil / D19-0	D19-XXX	D-Series Base Oil / D30-0	C24-XXX	C-Series Base Oil
Aluminum (Al)		1.00 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Chromium (Cr)		1.00 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Copper (Cu)		1.00 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Iron (Fe)		1.00 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Lead (Pb)		1.00 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Magnesium (Mg)		1.00 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Nickel (Ni)		1.00 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Silicon (Si)		1.00 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Silver (Ag)		1.00 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Sodium (Na)		1.00 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Tin (Sn)		1.00 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Titanium (Ti)		0.92 X		<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Boron (B)	1.15 X			<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Molybdenum (Mo)	0.85 X			<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Zinc (Zn)	1.00 X			<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Barium (Ba)			1.00 X	<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Cadmium (Cd)			1.00 X	<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Manganese (Mn)			1.00 X	<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Vanadium (V)			1.00 X	<1 ppm	1.00 X	<1 ppm	1.00 X	<1 ppm
Calcium (Ca)			1.00 X			<1 ppm	1.00 X	<1 ppm
Potassium (K)			1.00 X			<1 ppm	1.00 X	<1 ppm
Lithium (Li)			1.00 X			<1 ppm	1.00 X	<1 ppm
Phosphorus (P)			1.00 X			<1 ppm	1.00 X	<1 ppm
Arsenic (As)			1.00 X			<1 ppm		
Bismuth (Bi)			1.00 X			<1 ppm		
Cerium (Ce)			1.00 X			<1 ppm		
Cobalt (Co)			1.00 X			<1 ppm		
Indium (In)			1.00 X			<1 ppm		
Tungsten (W)			1.00 X			<1 ppm		
Zirconium (Zr)			1.00 X			<1 ppm		
Antimony (Sb)							1.00 X	<1 ppm

Where X = nominal concentration of spectrometric oil standard in ppm

3.3.2 Type D12-XXX. Type D12-XXX spectrometric oil standard shall meet the Metal Elements (RDE-AES and ICP-AES) requirements in table II and contain the elements at the concentration as listed in table IV.

3.3.3 Type D15X-XXX. Type D15X-XXX spectrometric oil standard shall meet the Metal Elements (RDE -AES and ICP-AES) requirements specified in table II and contain the elements at the concentration as listed in table IV.

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3.3.4 Type D-Series base oil/D19-0. Type D-Series base oil/D19-0 shall be a zero (0) standard that meets all of the requirements specified in table II and does not exceed the elemental requirements as listed in table IV.

3.3.5 Type D19-XXX. Type D19-XXX spectrometric oil standard shall meet the Metal Elements (RDE -AES and ICP-AES) requirements specified in table II and contain the elements at the concentration as listed in table IV.

3.3.6 Type D-Series base il/D30-0. Type D-Series base oil/D30-0 shall be a zero (0) standard that meets all of the requirements specified in table II and does not exceed the elemental requirements as listed in table IV.

3.3.7 Type C24-XXX. Type C24-XXX spectrometric oil standard shall meet the Metal Elements (RDE -AES and ICP-AES) requirements specified in table III and contain the elements at the concentration as listed in table IV.

3.4 Homogeneity. All blended spectrometric oil standards shall be homogeneous, free of sediment or precipitate, and shall not indicate any mirroring or silvering appearance from the silver element.

3.5 Spectrometric oil standards stability. The spectrometric oil standards shall have a storage stability of not less than 30 months for Types D12-XXX, D19-0, and D30-0 and not less than 12 months for Types D3-XXX, D15X-XXX, D19-XXX and C24-XXX.

3.6 Toxicity. The spectrometric oil standards shall have no adverse effect on the health of personnel when used for its intended purpose in accordance with the safety standards issued by the Occupational Safety and Health Administration (OSHA) (see 4.2.1, 6.6 and 6.7).

3.7 Properties. Base oils shall meet all of the requirements of this specification when tested as specified in table II or table III, and section 4. Spectrometric oil standards shall meet the Metal Elements (RDE -AES and ICP-AES) requirements of this specification when tested as specified in table II or table III, and section 4

3.8 NIST Traceability of Standards. All calibration standards and reference materials used to meet the requirements of this specification shall be traceable to the appropriate Standard Reference Materials (SRMs) maintained by the National Institute of Standards and Technology (NIST).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. Qualification inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall consist of testing to all the tests specified in table II or table III, and 4.4. The storage stability test (see 3.5, 4.4.3) shall be performed after the candidate product has passed all other qualification tests. Tentative qualification approval may be granted to products undergoing the storage stability tests. Upon successful completion of the storage stability test, full qualification approval may be granted. Failure to pass the storage stability test is cause for withdrawal of qualification approval.

4.2.1 Toxicity. The spectrometric oil standards shall be evaluated by the Navy and Marine Corps Force Health Protection Command (NMCFHPC), using the administrative Health Hazard Assessment (HHA) (see 6.7).

4.2.2 Sampling. Qualification samples shall consist of eight 8 ounce bottles of type D19-XXX in nominal concentrations of 5 ppm, 10 ppm, 30 ppm, 50 ppm, 80 ppm, 100 ppm, 120 ppm, 300 ppm, 500 ppm, 700 ppm, and 900 ppm; eight 8 ounce bottles of type D15X-XXX in the nominal concentrations of 5 ppm, 10 ppm, 30 ppm, 50 ppm, 100 ppm; eight 8 ounce bottles of type D12-XXX in nominal concentrations of 5 ppm, 10 ppm, 30 ppm, 50 ppm, 100 ppm, and 300 ppm; eight 8 ounces bottles of type D3-100; eight 8 ounce bottles of type D19-0; eight 8 ounce bottles of type D30-0; eight 8 ounce bottles of type C24-100 and eight 8 ounce bottles of C-Series base oil (see 6.3.)

4.3 Conformance inspection. Conformance inspection shall consist of the tests specified in table V and table VI. Failure to pass any conformance test is cause for rejection.

Table V. Conformance tests base oil.

Physical property	Requirement	Test Method
Viscosity (mm^2 /s at 100 °C)	1/	ASTM D445 ASTM D7042
Metal Elements (ICP-AES)	Meets accuracy and repeatability as outlined in paragraph 4 for the elements in D-Series Base Oil / D19-0, D-Series Base Oil / D30-0, and C-Series Base Oil respectively	ASTM D5185

1/ Values are required and used as comparative reference for testing.

Table VI. Conformance tests standards.

Physical property	Requirement	Test Method
Metal Elements (ICP-AES)	Meets accuracy and repeatability as outlined in paragraph 4.4 for the elements for each Spectrometric Oil Standard Type	ASTM D5185

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4.3.1 Sampling and inspection. Each bulk lot (see 6.5) of material for conformance inspection shall be sampled at random in accordance with ASTM D4057 or ASTM D4177.

4.3.2 Examination of filled containers. A random sample of filled containers from each lot (see 6.5), taken in accordance with ASQ/ANSI Z1.4, shall be examined with regard to fill, closure, sealing and leakage. Reject any container having one or more defects or for being under the required fill.

4.3.3 Conformance inspection. The conformance tests listed in Table V and Table VI shall be performed by the manufacturer for review and approval by the qualifying activity on each production run produced for the U.S. Government prior to acceptance (see 6.2 and 6.4).

4.3.4 Inspection conditions. All inspections shall be performed in accordance with the test conditions specified herein.

4.4 Test methods. All tests shall be performed in accordance with table II and table III and 4.4.1 through 4.4.3. Failure to pass any test is cause for rejection.

4.4.1 RDE-AES qualification testing. Testing of spectrometric oil standards shall be done as a statistical analysis and performed on the test results generated by the spectrometric analysis to provide an accuracy index (AI) and repeatability index (RI) for each element involved.

4.4.1.1 Test method. The qualification testing of spectrometric oil standards shall consist of ten replicates of each standard. The results shall be obtained with a Spectroil M/N spectrometer or equivalent direct reading, RDE-AES, approved by the Joint Oil Analysis Program-Coordinating Group (JOAP-CG) (See 6.4).

4.4.1.1.1 Accuracy index (AI) by Spectroil M/N. The accuracy index (AI) of spectrometric oil standards shall meet the requirements specified in table VII for the appropriate ppm concentration level. If the spectrometric oil standard does not meet the specified accuracy index, then the spectrometric oil standard shall be rejected.

The accuracy index is as follows:

$$AI = |Y - \bar{X}|$$

Where Y = nominal value of the spectrometric oil standard in ppm (see table IV)
$\bar{X}$ = average measured value of the spectrometric oil standard in ppm

4.4.1.1.2 Repeatability index (RI) by Spectroil M/N. The repeatability index (RI) of the spectrometric oil standards shall meet the requirements specified in table VII for the appropriate ppm concentration level. If the results are not satisfactory for the spectrometric oil standards, then the spectrometric oil standard shall be rejected.

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The repeatability index is as follows:

$$\sqrt{\frac{N(\sum X_i^2) - (\sum X_i)^2}{N(N-1)}}$$

Where N = number of replicates = 10

X_i= individual analysis result

Σ = summation of the analytical data

4.4.2 ICP qualification testing. Testing of spectrometric oil standards shall be done as a statistical analysis and performed on the data generated by the spectrometric analysis to provide an accuracy index (AI) and repeatability index (RI) for each element involved.

4.4.2.1 Test method. The qualification testing of spectrometric oil standards shall consist of three (3) replicates of each standard. Spectrometric oil standards shall be diluted on a weight-by-weight basis with a suitable ICP solvent. Typically, spectrometric oil standards in concentrations 5 ppm, 10 ppm, 30 ppm, and 50 ppm shall be diluted tenfold. Spectrometric oil standards in concentrations 80 ppm, 100 ppm, 120 ppm, 300 ppm, 500 ppm, 700 ppm, and 900 ppm shall be diluted to an appropriate concentration for the given calibration curve. The results shall be obtained with an ICP-AES approved by the Joint Oil Analysis Program-Coordinating Group (JOAP-CG)(see 6.4).

4.4.2.1.1 Accuracy index (AI) for ICP. The accuracy index (AI) of spectrometric oil standards shall meet the requirements specified in table VIII for the appropriate ppm concentration level. If the spectrometric oil standard does not meet the specified accuracy index, then the spectrometric oil standard shall be rejected.

The accuracy index is as follows:

$$AI = |Y - \bar{X}|$$

Where Y = formulation concentration of the spectrometric oil standard in ppm (see table VIII)

$\bar{X}$ = average measured value of the spectrometric oil standard in ppm

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Table VII. Metal elements (RDE-AES) accuracy and repeatability index.

	Accuracy Index (AI)												Repeatability Index (RI)											
Element	0	5	10	30	50	80	100	120	300	500	700	900	0	5	10	30	50	80	100	120	300	500	700	900
Al	0.88	1.20	1.59	3.33	5.12	7.93	9.65	11.55	27.80	46.00	64.20	82.40	0.50	0.56	0.71	1.58	2.55	4.17	5.03	6.16	15.00	25.00	35.00	45.00
Cr	0.88	1.20	1.59	3.33	5.12	7.93	9.65	11.55	27.80	46.00	64.20	82.40	0.50	0.56	0.71	1.58	2.55	4.17	5.03	6.16	15.00	25.00	35.00	45.00
Cu	0.92	1.61	2.44	5.91	9.43	14.79	18.20	21.83	53.50	88.80	124.00	159.00	0.50	0.56	0.71	1.58	2.55	8.27	5.03	11.92	27.00	45.00	63.00	81.00
Fe	0.91	1.50	2.21	5.23	8.29	12.97	16.00	19.11	46.70	77.50	108.00	139.00	0.50	0.56	0.71	1.58	2.55	7.15	5.03	10.38	24.00	40.00	56.00	72.00
Pb	1.60	1.98	2.43	4.47	6.64	10.13	12.20	14.56	34.30	56.60	78.80	101.00	0.90	0.95	1.08	2.01	3.13	5.16	6.07	7.53	18.00	30.00	42.00	54.00
Mg	0.92	1.61	2.44	5.91	9.43	14.79	18.20	21.83	53.50	88.80	124.00	159.00	0.50	0.56	0.71	1.58	2.55	6.32	5.03	9.97	27.00	45.00	63.00	81.00
Ni	0.88	1.20	1.59	3.33	5.12	7.93	9.56	11.55	27.80	46.00	64.20	82.40	0.50	0.56	0.71	1.58	2.55	4.17	5.03	6.16	15.00	25.00	35.00	45.00
Si	0.88	1.20	1.59	3.33	5.12	7.93	9.65	11.55	27.80	46.00	64.20	82.40	0.50	0.56	0.71	1.58	2.55	4.17	5.03	6.16	15.00	25.00	35.00	45.00
Ag	0.91	1.50	2.21	5.23	8.29	12.97	16.00	19.11	46.70	77.50	108.00	139.00	0.50	0.64	0.94	2.45	4.03	6.54	8.02	9.73	24.00	40.00	56.00	72.00
Na	1.01	2.59	4.36	11.60	18.90	29.90	37.10	44.45	110.00	183.00	255.00	328.00	0.50	0.64	0.94	2.45	4.03	6.40	8.02	9.47	48.00	80.00	112.00	144.00
Sn	1.60	1.98	2.43	4.47	6.64	10.13	12.20	14.56	34.30	56.60	78.80	101.00	1.00	1.04	1.17	2.06	3.16	5.21	6.08	7.58	18.00	30.00	42.00	54.00
Ti	0.89	1.30	1.78	3.93	6.14	9.54	11.70	13.99	33.90	56.10	78.30	101.00	0.50	0.58	0.78	1.87	3.04	4.96	6.02	7.35	18.00	30.00	42.00	54.00
B	0.89	1.30	1.78	3.93	6.14	9.54	11.70	13.99	33.90	56.10	78.30	101.00	0.50	0.58	0.78	1.87	3.04	4.96	6.02	7.35	18.00	30.00	42.00	54.00
Mo	0.91	1.50	2.21	5.23	8.29	12.97	16.00	19.11	46.70	77.50	108.00	139.00	0.50	0.64	0.94	2.45	4.03	6.54	8.02	9.73	24.00	40.00	56.00	72.00
Zn	0.96	1.99	3.19	8.15	13.10	20.72	25.60	30.74	75.60	126.00	176.00	226.00	0.50	0.78	1.30	3.63	6.02	9.72	12.00	14.52	36.00	60.00	84.00	108.00
Ba	0.50	0.72	1.50	3.25	5.50	8.63	10.00	12.84	32.00	53.00	73.98	95.00	0.50	0.75	0.75	1.75	2.75	3.22	3.75	4.94	10.00	20.00	29.82	40.00
Cd	0.50	0.72	1.50	3.25	5.50	8.63	10.00	12.84	32.00	53.00	73.98	95.00	0.50	0.75	0.75	1.75	2.75	3.22	3.75	4.94	10.00	20.00	29.82	40.00
Mn	0.50	0.72	1.50	3.25	5.50	8.63	10.00	12.84	32.00	53.00	73.98	95.00	0.50	0.79	0.75	1.75	2.75	3.32	3.75	4.66	12.00	25.00	31.74	40.00
V	0.50	0.72	1.50	3.25	5.50	8.63	10.00	12.84	32.00	53.00	73.98	95.00	0.50	0.79	0.75	1.75	2.75	3.32	3.75	4.66	10.00	20.00	32.84	45.00
Ca	0.50	0.72	1.50	3.25	5.50		10.00						0.50	0.75	0.75	1.75	2.75		3.75					
K	0.50	0.72	1.50	3.25	5.50		10.00						0.50	0.75	0.75	1.75	2.75		3.75					
Li	0.50	0.72	1.50	3.25	5.50		10.00						0.50	0.75	0.75	1.75	2.75		3.75					
P	10.16	10.67	11.18	15.50	18.20		20.00						5.02	5.21	5.41	5.50	7.50		10.00					
As	0.50	0.72	1.50	3.25	5.50		10.00						0.50	0.75	0.75	1.75	2.75		3.75					
Bi	0.50	0.72	1.50	3.25	5.50		10.00						0.50	0.75	0.75	1.75	2.75		3.75					
Ce	0.50	0.72	1.50	3.25	5.50		10.00						0.50	0.75	0.75	1.75	2.75		3.75					
Co	0.50	0.72	1.50	3.25	5.50		10.00						0.50	0.75	0.75	1.75	2.75		3.75					
In	0.50	0.72	1.50	3.25	5.50		10.00						0.50	0.75	0.75	1.75	2.75		3.75					
W	0.50	0.72	1.50	3.25	5.50		10.00						0.50	0.75	0.75	1.75	2.75		3.75					
Zr	0.50	0.72	1.50	3.25	5.50		10.00						0.50	0.75	0.75	1.75	2.75		3.75					

Note: The column headings, 0, 5, 10, 30, 50, 80, 100, 120, 300, 500, 700 and 900 denote the nominal concentration in parts per million (ppm).

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4.4.2.1.2 Repeatability index (RI) for ICP. The repeatability index (RI) of the spectrometric oil standards shall meet the requirements specified in table VIII for the appropriate ppm concentration level. If the results are not satisfactory for the spectrometric oil standards, then the spectrometric oil standard shall be rejected.

The index is as follows repeatability:

$$\sqrt{\frac{N(\sum X_i^2) - (\sum X_i)^2}{N(N-1)}}$$

Where N = number of replicates = 3

X_i= individual analysis result

Σ = summation of the analytical data

4.4.3 Storage stability test. Two 8-ounce samples for each type of spectrometric oil standards shall be stored at room temperature maintained at 25 ±5 °C for the following periods. Type D3-XXX: 12 months, Type D12-XXX: 30 months, Type D15X-XXX: 12 months, Type D19-0: 30 months, Type D19-XXX: 12 months, Type D30-0: 30 months, Type C24-XXX: 12 months. At the midpoint and end of these storage periods, the respective sample shall be examined for conformance to the requirements specified in table V and table VI.

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Table VIII: ICP-AES (ASTM D5185) accuracy and repeatability index.

	Accuracy Index (AI)																			Repeatability Index (RI)																			
	Type D19-0, D30-0, and C-Series Base Oil	Type D15X-XXX, Type D19-XXX, and Type C24-100												Type D12-XXX						Type D3-100		Type D19-0, D30-0, and C-Series Base Oil	Type D15X-XXX, Type D19-XXX, and Type C24-100												Type D12-XXX						Type D3-100
Element	0	5	10	30	50	80	100	120	300	500	700	900	5	10	30	50	100	300	100	Element	0	5	10	30	50	80	100	120	300	500	700	900	5	10	30	50	100	300	100
Al	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5	1.0	3.0	5.0	10.0	30.0		Al	1.0	1.4	1.8	2.9	3.5	4.3	4.7	5.1	7.4	9.1	10.4	11.5	1.4	1.8	2.9	3.5	4.7	7.4	
Cr	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5	1.0	3.0	5.0	10.0	30.0		Cr	1.0	0.6	1.0	2.2	3.2	4.5	5.4	6.2	12.3	18.0	23.1	27.9	0.6	1.0	2.2	3.2	5.4	12.3	
Cu	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5	1.0	3.0	5.0	10.0	30.0		Cu	1.0	0.5	1.0	2.7	4.2	6.5	7.9	9.4	21.5	34.3	46.6	58.6	0.5	1.0	2.7	4.2	7.9	21.5	
Fe	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5	1.0	3.0	5.0	10.0	30.0		Fe	1.0	0.5	0.8	2.0	3.0	4.3	5.2	6.0	12.5	18.8	24.5	30.0	0.5	0.8	2.0	3.0	5.2	12.5	
Pb	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5	1.0	3.0	5.0	10.0	30.0		Pb	1.0	2.7	3.3	4.8	5.6	6.5	7.0	7.4	9.9	11.7	13.0	14.1	2.7	3.3	4.8	5.6	7.0	9.9	
Mg	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5	1.0	3.0	5.0	10.0	30.0		Mg	1.0	0.6	1.2	3.0	4.6	6.9	8.4	9.8	21.6	33.5	44.8	55.6	0.6	1.2	3.0	4.6	8.4	21.6	
Ni	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5	1.0	3.0	5.0	10.0	30.0		Ni	1.0	1.1	1.6	2.8	3.5	4.5	5.0	5.4	8.5	10.9	12.9	14.6	1.1	1.6	2.8	3.5	5.0	8.5	
Si	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5	1.0	3.0	5.0	10.0	30.0		Si	1.0	2.0	2.4	3.1	3.6	4.1	4.3	4.5	5.7	6.5	7.1	7.6	2.0	2.4	3.1	3.6	4.3	5.7	
Ag	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5	1.0	3.0	5.0	10.0	30.0		Ag	1.0	0.6	1.0	2.5	3.9	5.7	6.9	8.0	17.1	26.1	34.5	42.5	0.6	1.0	2.5	3.9	6.9	17.1	
Na	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5	1.0	3.0	5.0	10.0	30.0		Na	1.0	1.4	2.2	4.6	6.5	8.8	10.2	11.5	21.1	29.6	37.0	43.7	1.4	2.2	4.6	6.5	10.2	21.1	
Sn	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5	1.0	3.0	5.0	10.0	30.0		Sn	1.0	3.2	3.5	4.3	4.7	5.1	5.3	5.4	6.3	6.9	7.3	7.6	3.2	3.5	4.3	4.7	5.3	6.3	
Ti	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0	0.5[A]	0.9[A]	2.8[A]	4.6[A]	9.2[A]	27.6[A]		Ti	1.0	1.0	1.3	1.9	2.3	2.7	3.0	3.2	4.5	5.4	6.1	6.7	0.9[A]	1.2[A]	1.8[A]	2.2[A]	2.9[A]	4.3[A]	
B	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0							11.5[B]	B	1.0	1.3	2.6	7.8	13.0	20.8	26.0	31.2	78.0	130.0	182.0	234.0							29.9[B]
Mo	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0							8.5[C]	Mo	1.0	0.9	1.5	3.1	4.5	6.2	7.3	8.3	15.7	22.5	28.4	33.9							6.5[C]
Zn	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0							10.0	Zn	1.0	0.6	1.1	3.0	4.7	7.1	8.6	10.1	22.7	35.6	47.8	59.7							8.6
Ba	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0								Ba	1.0	0.7	1.1	2.3	3.2	4.3	5.0	5.7	10.4	14.5	18.1	21.4							
Cd	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0								Cd	1.0	0.8	1.5	4.5	7.5	12.0	15.0	18.0	45.0	75.0	105.0	135.0							
Mn	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0								Mn	1.0	0.1	0.2	0.8	1.6	3.0	4.0	5.0	16.6	32.3	50.0	69.3							
V	1.0	0.5	1.0	3.0	5.0	8.0	10.0	12.0	30.0	50.0	70.0	90.0								V	1.0	0.3	0.6	1.8	3.1	4.9	6.1	7.3	18.3	30.5	42.7	54.9							
Ca	1.0	0.5	1.0	3.0	5.0		10.0													Ca	1.0	0.8	1.5	4.5	7.5		15.0												
K	1.0	0.5	1.0	3.0	5.0		10.0													K	1.0	0.8	1.5	4.5	7.5		15.0												
Li	1.0	0.5	1.0	3.0	5.0		10.0													Li	1.0	0.8	1.5	4.5	7.5		15.0												
P	1.0	0.5	1.0	3.0	5.0		10.0													Pb	1.0	0.8	1.5	4.5	7.5		15.0												
As	1.0	0.5	1.0	3.0	5.0		10.0													As	1.0	0.8	1.5	4.5	7.5		15.0												
Bi	1.0	0.5	1.0	3.0	5.0		10.0													Bi	1.0	0.8	1.5	4.5	7.5		15.0												
Ce	1.0	0.5	1.0	3.0	5.0		10.0													Ce	1.0	0.8	1.5	4.5	7.5		15.0												
Co	1.0	0.5	1.0	3.0	5.0		10.0													Co	1.0	0.8	1.5	4.5	7.5		15.0												
In	1.0	0.5	1.0	3.0	5.0		10.0													Ln	1.0	0.8	1.5	4.5	7.5		15.0												
W	1.0	0.5	1.0	3.0	5.0		10.0													W	1.0	0.8	1.5	4.5	7.5		15.0												
Zr	1.0	0.5	1.0	3.0	5.0		10.0													Zr	1.0	0.8	1.5	4.5	7.5		15.0												

Note: The column headings, 0, 5, 10, 30, 50, 80, 100, 120, 300, 500, 700, and 900 denote the nominal concentration in parts per million (ppm).

[A] - Ti concentration = 0.92X, where X = nominal concentration of spectrometric oil standard in ppm.

[B] - B concentration = 1.15X, where X = nominal concentration of spectrometric oil standard in ppm.

[C] - Mo concentration = 0.85X, where X = nominal concentration of spectrometric oil standard in ppm.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The spectrometric oil standards covered by this specification are intended for use in calibration and standardization of spectrometers used in spectrometric analysis for metallic elements found in oils and other fluids. These standards are intended for but limited to use in RDE-AES used by the Joint Oil Analysis Program (JOAP). JOAP has two (2) RDE-AES in use today, a spectrometer configured for fifteen (15) elements and a spectrometer configured for thirty (30) elements. The JOAP's formulations for spectrometric oil standards are specific to military hardware and logistics requirements.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type required (see 1.2)
c. Concentration required (see 3.3)
d. Quantity desired
e. Submittal of conformance test results (see 4.3.3)
f. Packaging requirements (see 5.1)
g. FAR clause 52.223-3

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-85694, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products (including the letter of authorization for submittal of samples) may be obtained from the Qualifying Activity Navy Oil Analysis Program Team Lead (AB44100), 22229 Elmer Road, Building 2360,

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Patuxent River, MD 20670. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Qualification submittal. To initiate the qualification process, forward a written request to submit product, to the qualification activity (see 6.3). A written response to this will be a "letter of authorization," to submit product for testing. The letter will provide detailed instructions for the submission of product (e.g., sample size, sample identification, testing fees, test data and reports, etc.).

6.3.2 Qualification approval. The qualifying activity upon completion of product testing will issue a “letter of notification” either notice of failure with testing progress or notice of product approval with a listing of mandatory manufacturer responsibilities to have their product listed and remain authorized on the Qualified Products List (QPL). As applicable, the approval letter will detail other stipulations of qualification, retention of qualification, QPL listing information, formal date of qualification, etc.

6.3.3 Retention of qualification. In order to retain qualification of a product approved for listing on the Qualified Product List (QPL), the Government will require the manufacturer to verify by certification to the qualifying activity that the manufacturer’s product complies with the requirements of this specification. The time of periodic verification by certification will be in two year intervals from the date of original qualification and will be initiated by the Government. The Government reserves the right to re-examine the qualified product whenever deemed necessary to ensure that the product continues to meet any or all of the specification requirements.

6.3.4 Miscellaneous. Spectrometric oil standards supplied under contract are to be identical in every respect to the samples tested and found to meet the requirements of this specification. Any unapproved changes from the qualification sample should constitute cause for rejection for material submitted and for removal from the list of qualified products. Accordingly, contact the Qualifying Activity for all anticipated changes to the product qualified prior to any production.

6.4 Joint Oil Analysis Program Coordinating Group (JOAP-CG). JOAP-CG can be contacted through the Navy Oil Analysis Program Team Lead, AB44100, 22229 Elmer Road, Patuxent River, MD 20670) or at noapsupport@us.navy.mil.

6.5 Definitions.

6.5.1 Bulk lot. A bulk lot is an indefinite quantity of homogeneous mixture of material offered for acceptance in a single isolated container or manufactured by a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

6.5.2 Parts per Million by Mass (ppm). A dimensionless unit of concentration representing one unit of mass of a constituent element for every one million units of mass of the total solution or mixture. For the purpose of this specification, the term ppm is used to represent the metric unit

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milligrams per kilometer (mg/kg). All concentrations values are reported on a mass -to-mass basis.

6.6 Toxicological data and formulations. The manufacturer should furnish the toxicological and formulation data to evaluate the safety of the material for the proposed use. A statement of the complete formulation of the product, to include the chemical name and percent of each ingredient, as well as a material safety data sheet of the product, prepared in accordance with FED-STD-313, should be submitted to the qualifying activity. The pertinent government mailing addresses for submission of data are listed in FED-STD-313.

6.7 Health/hazard assessment (HHA). The qualifying activity will request the Navy and Marine Corps Force Health Protection Command (NMCFHPC) to perform an HHA on the barrier material furnished. Prior to listing on the Qualified Products List, the results of the NMCFHPC HHA must be deemed acceptable by the qualifying activity. A flowchart for the HHA process can be found as enclosure (1) of BUMEDINST 6270.8. Additionally, information can be found on the website: https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquistion-Technical-Support/Health-Hazard-Assessments-HHA/. The HHA is a review of the product based on information submitted by the manufacturer, to assess health hazards associated with the handling, application, use and removal of the product. In order to process the HHA, NMCFHPC will request the manufacturer to provide sufficient information be sent to Commander, the Navy and Marine Corps Force Health Protection Command, Industrial Hygiene – HHA, 620 John Paul Jones Circle, Suite 1100, Portsmouth, VA 20378-2130 (see 4.2.1).

6.8 Subject term (key word) listing.

Calibration
Metal content in oil and other fluids
Metallo-organic concentrates
Spectrometer calibration
Wear Metals

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – AS
Air Force – 68
DLA – GS

Review activities:
Army – AM, AV
Air Force – 20

Preparing activity:
Navy – AS

(Project 9150-2025-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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